A Complete Algorithmic Toolbox for Congressional Redistricting: Legal Requirements, Algorithm Components, and Callais-Compliant Design for the Post-2026 Redistricting Cycle

B.0 Series Synthesis

Gio Della-Libera
Apportionment Research Group
`giodl@microsoft.com`

May 2026

Abstract

Congressional redistricting must satisfy eleven distinct legal requirements spanning constitutional population equality, geographic compactness, county preservation, the Voting Rights Act, and the racial-predominance bar set by *Shaw v. Reno* (1993). The 2026 Supreme Court decision in *Louisiana v. Callais* adds a critical constraint: race-conscious and partisan signals must be kept *disentangled* in any redistricting algorithm—they cannot be combined in a single optimisation run.

We present a unified algorithmic toolbox organised along three axes: (A) vertex weights governing what METIS balances, (B) edge management controlling compactness and subdivision preservation, and (C) division strategy determining how the bisection tree is structured. Fifteen algorithm components developed across the B-series papers (B.1–B.15) map to these axes, and each legal requirement maps to at least one component.

The toolbox is structured so that each mode uses exactly one signal type. **VRA-Section** (B.14) addresses Gingles Prong 1 using geographic alignment alone—no racial target, no racial count. **ProportionalSection** (B.12, planned) addresses proportionality using partisan vote share alone. A `callais_preflight` gate enforces mutual exclusion: the two signals cannot be combined in one run. This is not a limitation but the legally required architecture under *Louisiana v. Callais*.

We present a bakeoff of eight algorithm combinations across four legally contested states (Wisconsin 8D, North Carolina 14D, Georgia 14D, Alabama 7D), measuring edge cut, population balance, D-seat count, majority-minority district count, natural bisection ratio, and county split count. The results confirm that geographic voter sorting—not the bisection algorithm— drives partisan outcomes in all four states, and

that VRASection produces the minority-opportunity district patterns required by *Allen v. Milligan* (2023) without any racial targeting.

A court can audit exactly which signals were used in any plan by reading the `whatif-manifest v1` JSON sidecar produced by the `redist` CLI. Every run is parameter-reproducible and hash-chain verifiable.

Contents

1 Introduction

1.1 The Redistricting Problem

Every ten years, following the decennial census, all 50 American states must redraw their congressional district boundaries. The process allocates 435 seats across states via the Huntington-Hill method (Article I, §2, operationalised since 1941), then asks each state to partition its population into exactly k districts where k is its apportioned seat count. The constitutional floor is simple: districts must be as close to equal in population as practicable (*Wesberry v. Sanders*, 1964, one person one vote). Everything above that floor has accumulated as a dense forest of legal requirements over six decades of redistricting litigation.

The forest has grown denser since 2020. *Rucho v. Common Cause* v. *Common Cause* (2019) removed federal courts from partisan gerrymandering review, but *immediately redirected challengers to state courts* under state constitutional grounds. Pennsylvania's *League of Women Voters* (2018) and North Carolina's *Harper v. Hall* (2022) established that state constitutions can ban partisan gerrymandering even where Rucho closes the federal door. *Allen v. Milligan* (2023) reaffirmed and clarified the Voting Rights Act §2 obligation to draw minority-opportunity districts where Gingles conditions are met. And *Louisiana v. Callais* (2026) settled the disentanglement question: a state may not use race-conscious criteria to justify what is actually partisan manipulation, and the "strong-inference" framework requires challengers to show that a state's stated political goals could have been achieved *with better minority outcomes*—meaning the algorithm must be capable of separating the two signals.

This paper provides what the litigation landscape demands: a *toolbox* of algorithm components where each tool addresses exactly one legal requirement, and the tools compose without conflict because they are legally required to be separate.

1.2 Why a Toolbox, Not a Single Algorithm

No single redistricting algorithm can satisfy all requirements simultaneously. The requirements pull in different directions. Geographic compactness (round districts) conflicts with county preservation (keeping existing political lines). VRA minority-opportunity district formation requires concentrating minority population, which can conflict with equal geographic spread. Proportionality requires a partisan signal, which *Louisiana v. Callais* prohibits from being mixed with a racial signal. Census stability across decades requires that the algorithm converges on the same geographic structure regardless of which census year's data is used.

The toolbox architecture acknowledges these tensions explicitly. Each tool is a parameter dial: the *geographic edge weights* of B.2 control compactness; the *county-sticky weights* of B.10 control subdivision preservation; the *VRASection alignment score* of B.14 controls Gingles Prong 1 alignment; the *AreaSection dual constraint* of B.9 controls geographic fairness. These dials can be tuned independently. They cannot all be maximised simultaneously, but the tradeoffs are *explicit and auditable*: a court can read the parameter values from the plan manifest and understand exactly which tradeoffs were made.

1.3 The Callais Constraint

Louisiana v. Callais, 608 U.S. ___ (2026), has three holdings relevant to algorithmic redistricting.

First, Louisiana's extra minority district was unconstitutional racial gerrymandering: the VRA did not require a second majority-minority district where the first compactness prong of *Thornburg v. Gingles* (1986) was not met.

Second, §2 challengers must use the "strong-inference" framework: they must show that the state's *stated political goals* (partisan advantage, community preservation, compactness) could have been achieved *with better minority outcomes.* This shifts the burden from "the map is racially unfair" to "a race-neutral algorithm could have done better on both dimensions". The toolbox is precisely positioned to provide this counterfactual: one can run GeoSection (no minority signal) and VRASection (geographic minority alignment) on the same state and compare.

Third, at p. 36, the majority holds that race-conscious and partisan signals must be *disentangled*: a redistricting run that mixes minority VAP concentrations as a constraint alongside partisan vote share as a constraint is constitutionally suspect because a court cannot separate the racial motivation from the partisan motivation in the resulting map.

This third holding is the architecturally decisive constraint. It means that a redistricting system with multiple simultaneous signals— minority population fractions plus partisan vote shares in the same METIS objective function—fails the Callais disentanglement test. The toolbox's structure, where each mode uses exactly one signal type, is therefore not a simplification or limitation but a constitutionally correct design.

1.4 Paper Scope and Contributions

This paper is the capstone of the B-series algorithmic redistricting programme. It synthesises findings from B.1 through B.15 into a unified framework.

Contributions.

1. A complete mapping of all eleven legal redistricting requirements to specific algorithm components (Section **??**).
2. A three-axis taxonomy of algorithm variants (Section **??**).
3. A bakeoff of eight algorithm combinations across four litigated states, with empirical results on compactness, partisanship, minority representation, and county preservation (Section **??**).
4. A formal Callais compliance architecture showing how the toolbox enforces p. 36 disentanglement at the CLI level (Section **??**).
5. Practitioner recommendations for which configuration to choose for which legal context (Section **??**).

Notation. Throughout this paper, $G = (V, E, w)$ denotes the state's census-tract adjacency graph with vertex population weights and edge boundary-length weights. k denotes the state's congressional seat count. $\mathrm{EC}(i : j)$ denotes the minimum edge-cut at split ratio $i : j$. $\mathrm{EC}_{\mathrm{norm}}(i : j) = \mathrm{EC}(i : j)/\sqrt{\min(i, j)}$ denotes the isoperimetrically normalised edge-cut introduced in GeoSection (B.8). $p^* = L(0.5)$ denotes the Lorenz-curve natural population fraction (population in the densest 50 % of land area). D votes % is the 2020 presidential Democratic two-party vote share.